S.M.A.R.T. Drive Triage

Testing and Results — Evidence Appendix

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I tested synthetic reports to check extraction, interpretation, assessment categories, recommendation ordering, and interface wording. I retained unsuccessful runs and repeated inputs as part of the development history. This appendix documents the submitted course-project version.

I include test definitions, observed output excerpts, individual results, and original output exhibits. I identify evidence by stable test and exhibit numbers rather than working filenames or communication timestamps.

Record basis

I distinguish a targeted pass from a complete output-quality pass. A correct category does not excuse an unsupported statement elsewhere. My final regression configuration was temperature 0.2 and maximum completion tokens 2048. Development print captures also show these values. The final cropped screenshots do not independently display every setting. I did not record a controlled comparison of parameter values.

I reconstructed 20 identifiable runs from retained results. I have kept earlier observations without a confirmed original-run mapping separate rather than inventing input details or counting them as additional recovered experiments.

Input definitions

Input A

SATA HDD: overall PASSED. Reallocated_Sector_Ct RAW_VALUE=24, VALUE=100, WORST=100, THRESH=010, Pre-fail. Current_Pending_Sector RAW_VALUE=8, VALUE=100, WORST=100, THRESH=000. Offline_Uncorrectable RAW_VALUE=8, VALUE=100, WORST=100, THRESH=000. Temperature_Celsius RAW_VALUE=35, VALUE=065, WORST=055, THRESH=000. The full original input appears in Exhibits E-01 to E-03.

Input B

NVMe SSD A: PASSED; Critical Warning 0x00; Temperature 35 Celsius; Available Spare 100%; Available Spare Threshold 10%; Percentage Used 3%; Power Cycles 150; Power On Hours 2400; Unsafe Shutdowns 0; Media and Data Integrity Errors 0; Error Information Log Entries 0; Warning Comp. Temperature Time 0; Critical Comp. Temperature Time 0.

Input C

NVMe SSD B: FAILED, available spare fallen below threshold; Critical Warning 0x01; Temperature 35 Celsius; Available Spare 5%; Available Spare Threshold 10%; Percentage Used 85%; Power Cycles 450; Power On Hours 12000; Unsafe Shutdowns 0; Media and Data Integrity Errors 12; Error Information Log Entries 12; both temperature-time counters 0. The changed-value retest uses spare 4% and both error counts 15.

Input D

SATA HDD C: PASSED; Reallocated_Sector_Ct raw 0, VALUE=100, WORST=100, THRESH=010, Pre-fail; Power_On_Hours raw 3200, VALUE=095, WORST=095, THRESH=000; Current_Pending_Sector raw 6 and Offline_Uncorrectable raw 6, both VALUE=100, WORST=100, THRESH=000. No previous reports or self-test results are supplied.

Input E

SATA HDD D: PASSED; Raw_Read_Error_Rate RAW_VALUE=987654321, VALUE=117, WORST=099, THRESH=006, Pre-fail. No vendor-specific raw-value encoding is supplied. No other attributes, previous reports, or self-test results are supplied.

Input F

NVMe SSD E: PASSED; Critical Warning 0x00; Temperature 34 Celsius; Available Spare 100%; threshold 10%; Percentage Used 4%; Power Cycles 175; Power On Hours 2600; Unsafe Shutdowns 0; both error counters 0; both temperature-time counters 0.

Input G

NVMe SSD F: FAILED, available spare fallen below threshold; Critical Warning 0x01; Temperature 34 Celsius; Available Spare 6%; threshold 10%; Percentage Used 82%; Power Cycles 425; Power On Hours 11500; Unsafe Shutdowns 0; Media and Data Integrity Errors 9; Error Information Log Entries 9; both temperature-time counters 0.

Input H

No S.M.A.R.T. information is available for this device.

Input I

Write a short poem about the ocean.

I summarize measurement content above for compact comparison. The following final-regression report definitions preserve the report structure and values with whitespace normalized for readability. Original development input captures are retained in the exhibits.

Full input F

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD E
NVMe Version: 1.4
=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED
SMART/Health Information (NVMe Log 0x02)
Critical Warning: 0x00
Temperature: 34 Celsius
Available Spare: 100%
Available Spare Threshold: 10%
Percentage Used: 4%
Power Cycles: 175
Power On Hours: 2600
Unsafe Shutdowns: 0
Media and Data Integrity Errors: 0
Error Information Log Entries: 0
Warning Comp. Temperature Time: 0
Critical Comp. Temperature Time: 0
```

Full input G

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD F
NVMe Version: 1.4
=== START OF SMART DATA SECTION ===
```

```
SMART overall-health self-assessment test result: FAILED!
- available spare has fallen below threshold
SMART/Health Information (NVMe Log 0x02)
Critical Warning: 0x01
Temperature: 34 Celsius
Available Spare: 6%
Available Spare Threshold: 10%
Percentage Used: 82%
Power Cycles: 425
Power On Hours: 11500
Unsafe Shutdowns: 0
Media and Data Integrity Errors: 9
Error Information Log Entries: 9
Warning Comp. Temperature Time: 0
Critical Comp. Temperature Time: 0
```

Full input D

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Device Model: SYNTHETIC SATA HDD C
Device type: ATA device
=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED
ID# ATTRIBUTE_NAME          FLAG    VALUE WORST THRESH TYPE      UPDATED WHEN_FAILED RAW_VALUE
5 Reallocated_Sector_Ct    0x0033 100   100   010    Pre-fail Always  -          0
9 Power_On_Hours           0x0032 095   095   000    Old_age  Always  -          3200
197 Current_Pending_Sector 0x0012 100   100   000    Old_age  Always  -          6
198 Offline_Uncorrectable 0x0010 100   100   000    Old_age  Offline -          6
No previous reports or self-test results are supplied.
```

Full input E

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Device Model: SYNTHETIC SATA HDD D
=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED
ID# ATTRIBUTE_NAME     FLAG    VALUE WORST THRESH TYPE      UPDATED WHEN_FAILED RAW_VALUE
1 Raw_Read_Error_Rate  0x000f 117   099   006    Pre-fail Always  -          987654321
No vendor-specific raw-value encoding is supplied.
No other attributes, previous reports, or self-test results are supplied.
```

Development run register

T-01 — Input A

Outcome: Partial pass.

Observed output: “the drive is deteriorating”; “imminent risk of data loss”; “This will identify any additional bad sectors”.

I obtained Urgent attention but rejected unsupported trend, timing, and diagnostic promises. A device-path command also appeared. I tightened the interpretation and action instructions.

Included evidence: Exhibit E-01.

T-02 — Input A

Outcome: Partial pass.

Observed output: “the drive still reports a healthy normalized value”; “confirm the extent”.

I observed improvement in the opening assessment but retained concerns about broad reassurance and diagnostic certainty.

Included evidence: Exhibit E-02.

T-03 — Input A

Outcome: Partial pass.

Observed output: “This will identify any additional bad sectors”; the device-path command returned.

I also observed dropped leading zeros. I moved displayed recommendations into fixed application text and strengthened value preservation.

Included evidence: Exhibit E-03.

T-04 — Input B

Outcome: Partial pass.

Observed output: “Only 3% of capacity used”; “moderate usage”.

I corrected endurance-versus-capacity interpretation and requested neutral descriptions of usage counters.

Included evidence: Exhibit E-04.

T-05 — Input B

Outcome: Partial pass.

Observed output: “Low endurance consumption”; “no indication of excessive cycling”.

I confirmed the endurance correction but kept counter neutrality open.

Included evidence: Exhibit E-05.

T-06 — Input C

Outcome: Fail: intended classification.

Observed output: “Monitor”; “Moderate operational hours”.

I added an explicit Urgent attention rule for failed health with below-threshold spare, plus a warning example.

Included evidence: Exhibit E-06.

T-07 — Input C, spare 4%, both error counts 15

Outcome: Targeted pass.

Observed output: “Urgent attention”; “Manufacturer-estimated endurance consumed; not occupied storage capacity.”

I confirmed changed values, separate error counts, neutral counters, and data protection first.

Included evidence: Exhibit E-07.

T-08 — Input B

Outcome: Partial pass.

Observed output: “Neutral metric; no indication of abnormal usage.”

I confirmed Important Notes, the category, and endurance interpretation, but rejected the unsupported usage judgment.

Included evidence: Exhibit E-08.

T-09 — Input B

Outcome: Partial pass: separate repeat.

Observed output: “Neutral metric; no indication of abnormal usage.”

I repeated the same input and observed the same defect. I replaced general guidance with specific descriptions.

Included evidence: Exhibit E-09.

T-10 — Input B

Outcome: Partial pass.

Observed output: “Reported number of power cycle events”; “Reported warning critical temperature time.”

I confirmed neutral usage descriptions but found mixed temperature labels.

Included evidence: Exhibit E-10.

T-11 — Input B

Outcome: Targeted pass.

Observed output: “Reported warning temperature time”; “Reported critical temperature time”.

I retained this unchanged-input repetition separately. Both temperature counters were distinct.

Included evidence: Exhibit E-11.

T-12 — Input B

Outcome: Partial pass.

Observed output: “Reported warning critical temperature time.”

I observed recurrence and added distinct warning/critical counter wording.

Included evidence: Exhibit E-12.

T-13 — Input B

Outcome: Partial pass.

Observed output: “Reported warning-temperature time”; “when the recorded errors occurred”.

I confirmed the temperature correction but kept the zero-error note relevance issue open. I later consolidated the scope and uncertainty rules.

Included evidence: Exhibit E-13.

T-14 — Input H: missing-telemetry check

I obtained Insufficient information, no usable measurements, general report/documentation guidance, and the note “Missing telemetry alone does not establish a hardware fault.” I observed no commands, device paths, or replacement advice. Outcome: targeted pass. I retain the result as a text record here.

Final regression register

T-15 — Input D

Expected and observed category: Urgent attention. Outcome: pass against the defined checks.

I checked separate counts, no unsupported trend or timing, no wear inference from hours or Pre-fail, no reassurance from PASSED, data protection first, no commands, and no diagnostic guarantees. All eight checks passed.

Observed excerpt: “Reported uncorrectable sectors; these may overlap with the pending sectors.”

Included visual evidence: Exhibit E-14.

T-16 — Input E

Expected and observed category: Insufficient information. Outcome: pass against the defined checks.

I preserved the raw number without a literal failed-read count or wear diagnosis. I observed no failure prediction, no Pre-fail misclassification, no reliability guarantee, and no replacement, scan, or command advice. All six checks passed.

Observed excerpt: “Not interpretable as a literal failed-read count without vendor documentation.”

T-17 — Input F

Expected and observed category: No obvious warning in supplied data. Outcome: pass against the defined checks.

I confirmed classification, endurance meaning, neutral cycles/hours, distinct temperature counters, bounded claims, and no irrelevant error-timing note. All six checks passed.

Observed excerpt: “Manufacturer-estimated endurance consumed.”

T-18 — Input G

Expected and observed category: Urgent attention. Outcome: pass against the defined checks.

I confirmed the category, 6% below 10%, endurance meaning, two counts of 9 kept separate, no unsupported trends/timing/judgments, and data protection first without commands or guarantees. All six checks passed.

Observed excerpt: “Reported spare capacity is below the supplied 10% threshold.”

T-19 — Input H

Expected and observed category: Insufficient information. Outcome: pass against the defined checks.

I confirmed the category, no invented measurements, no invented fault, general report/support guidance, and no commands or replacement advice. All five checks passed.

Observed excerpt: “Missing telemetry alone does not establish a hardware fault.”

Included visual evidence: Exhibit E-15.

T-20 — Input I

Expected and observed category: Invalid input. Outcome: pass against the defined checks.

I confirmed the category, no invented telemetry or poem, and a request for an appropriate report. All three checks passed.

Observed excerpt: “Please supply a S.M.A.R.T. report rather than a general request.”

Complete retained unrelated-input output

```
Triage assessment
Invalid input
Please supply a S.M.A.R.T. report rather than a general request.
Evidence
No usable measurements were extracted.
Recommended actions
Paste a S.M.A.R.T. report into the input box.
Important Notes
No drive telemetry was supplied.
This AI assessment can be wrong and cannot guarantee drive reliability.
```

Small checks and evidence coverage

I checked placement of the NVMe and priority instructions within SYSTEM_PROMPT. I checked the visible Important Notes heading independently of the internal limitations key, and confirmed the heading in developmental and final outputs. I used changed values to check application beyond literal reproduction of

examples. I kept identifiable unchanged-input repetitions separate.

I retain three additional historical observations: broad normal-range wording, surface-scan advice before backup in a limited-evidence case, and an imprecise wear interpretation. Their original input/run mappings remain unconfirmed. I have not invented entries or counted them as separately recovered experiments.

The following exhibits contain the available original development captures and selected regression captures. Other final cases are represented by input definitions, observed excerpts, and recorded results above; these excerpts are not complete verbatim transcripts. Original print headers remain in the unaltered captures but are not acceptance criteria.

Assessment Scope

I completed six final cases and each met its defined checks. This is a small manual regression set, not a general accuracy measurement or a guarantee of future consistency. Some cases resemble the prompt examples. Earlier repeated runs used preceding revisions rather than the consolidated final prompt. The final set does not establish Monitor-category coverage, broad manufacturer coverage, or adversarial-input robustness.

I interpret supplied reports as snapshots rather than evidence of historical deterioration or exact failure timing. I preserve unexplained vendor-specific values rather than assign undocumented meanings. Structural validation checks data types and known categories, not every factual claim. Fixed recommendations control their wording but depend on the model selecting an appropriate category. This appendix records the submitted project; it does not commit to further development.

Original output exhibits

I identify each capture by exhibit number. The evidence is included below, not left as references to unattached working files.

Exhibit E-01

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

S.M.A.R.T. report

```
smartctl 7.4
Device Model: SYNTHETIC TEST HDD
Device type: SATA HDD

SMART overall-health self-assessment test result: PASSED

ID# ATTRIBUTE_NAME          FLAG     VALUE WORST THRESH TYPE      UPDATED  WHEN_FAILED RAW_VALUE
  5 Reallocated_Sector_Ct   0x0033   100   100   010    Pre-fail  Always       -       24
197 Current_Pending_Sector  0x0012   100   100   000    Old_age   Always       -       8
198 Offline_Uncorrectable   0x0010   100   100   000    Old_age   Offline      -       8
194 Temperature_Celsius     0x0022   065   055   000    Old_age   Always       -       35
```

Analyze

Triage assessment

Urgent attention – The drive has 24 reallocated sectors, 8 pending sectors, and 8 uncorrectable sectors, all of which are warning-level indicators that the drive is deteriorating.

Evidence

Attribute	Reported value	Interpretation
Reallocated_Sector_Ct	RAW_VALUE=24, VALUE=100, WORST=100, THRESH=010	24 sectors have been remapped to spare sectors. This is a clear warning that the drive is experiencing physical defects.
Current_Pending_Sector	RAW_VALUE=8, VALUE=100, WORST=100, THRESH=000	8 sectors are currently marked as pending reallocation. They may become reallocated if read/write attempts fail.
Offline_Uncorrectable	RAW_VALUE=8, VALUE=100, WORST=100, THRESH=000	8 sectors were found uncorrectable during an offline scan. These sectors are already considered bad.
Temperature_Celsius	RAW_VALUE=35, VALUE=065, WORST=055, THRESH=000	35 °C is within normal operating temperature; not a concern.

Recommended actions

1. **Back up all critical data immediately.**
   The presence of reallocated, pending, and uncorrectable sectors indicates imminent risk of

data loss.

2. **Run a full surface scan (e.g., `smartctl -t long /dev/sdX`).**
   This will identify any additional bad sectors and confirm whether the current counts are stable or increasing.
3. **Plan for replacement.**
   If the surface scan reports more reallocated/pending sectors or if the counts grow over time, replace the drive as soon as possible.

Limitations

- The report is a single snapshot; no trend data is available to predict a failure date.
- Only a subset of SMART attributes is shown; other potentially relevant metrics (e.g., spin-up time, seek errors) are missing.
- The interpretation of raw values is based solely on the supplied data; vendor-specific encoding is not inferred.
- This assessment does not guarantee reliability; it only reflects the evidence present.

Exhibit E-02

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

S.M.A.R.T. report

```
smartctl 7.4
Device Model: SYNTHETIC TEST HDD
Device type: SATA HDD

SMART overall-health self-assessment test result: PASSED

ID# ATTRIBUTE_NAME          FLAG     VALUE WORST THRESH TYPE      UPDATED  WHEN_FAILED RAW_VALUE
  5 Reallocated_Sector_Ct   0x0033   100   100   010    Pre-fail  Always       -       24
197 Current_Pending_Sector  0x0012   100   100   000    Old_age   Always       -       8
198 Offline_Uncorrectable   0x0010   100   100   000    Old_age   Offline      -       8
194 Temperature_Celsius     0x0022   065   055   000    Old_age   Always       -       35
```

Analyze

Triage assessment

Urgent attention – The report shows non-zero Current Pending Sector and Offline Uncorrectable counts, which are typical early signs of sector-level problems.

Evidence

Attribute	Reported value	Interpretation
Reallocated_Sector_Ct	RAW = 24, VALUE = 100, WORST = 100, THRESH = 010	24 sectors have been reallocated; the drive still reports a healthy normalized value.
Current_Pending_Sector	RAW = 8, VALUE = 100, WORST = 100, THRESH = 000	8 sectors are pending reallocation; the drive has not yet reallocated them.
Offline_Uncorrectable	RAW = 8, VALUE = 100, WORST = 100, THRESH = 000	8 sectors were found uncorrectable during an offline scan.
Temperature_Celsius	RAW = 35 °C, VALUE = 065, WORST = 055, THRESH = 000	Current temperature is within normal limits.

Recommended actions

1. **Back up all critical data immediately.**
2. **Run a full surface scan** (e.g., using a tool that reports bad sectors) to confirm the extent of the pending and uncorrectable sectors.
3. **Plan for replacement** of the drive once the scan is complete, or keep a close watch on the

SMART attributes if replacement is not immediately possible.

Limitations

- The assessment is based on a single SMART snapshot; it does not show a trend over time.
- Raw values are reported exactly as supplied; we cannot infer the exact encoding or whether the counts are cumulative or per-scan.
- A “PASSED” overall-health status does not guarantee reliability when pending or uncorrectable sectors are present.
- Further diagnostics are needed to determine the exact scope of the problem.

Exhibit E-03

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information before submitting.

S.M.A.R.T. report

```
smartctl 7.4
Device Model: SYNTHETIC TEST HDD
Device type: SATA HDD

SMART overall-health self-assessment test result: PASSED

ID# ATTRIBUTE_NAME          FLAG     VALUE WORST THRESH TYPE      UPDATED  WHEN_FAILED
RAW_VALUE
  5 Reallocated_Sector_Ct   0x0033   100   100   010    Pre-fail  Always       -       24
197 Current_Pending_Sector  0x0012   100   100   000    Old_age   Always       -       8
198 Offline_Uncorrectable   0x0010   100   100   000    Old_age   Offline      -       8
194 Temperature_Celsius     0x0022   065   055   000    Old_age   Always       -       35
```

Analyze

Triage assessment

Urgent attention – The report shows non-zero counts of pending and uncorrectable sectors,

which are warning signs that the drive is experiencing ongoing errors.

Evidence

Attribute or finding	Reported value	Interpretation
Reallocated_Sector_Ct	RAW_VALUE = 24, VALUE = 100, THRESH = 10	24 sectors have already been moved to spare space.
Current_Pending_Sector	RAW_VALUE = 8, VALUE = 100, THRESH = 0	8 sectors are currently marked as pending; the drive cannot read them reliably.
Offline_Uncorrectable	RAW_VALUE = 8, VALUE = 100, THRESH = 0	8 sectors were found uncorrectable during an offline scan.
Temperature_Celsius	RAW_VALUE = 35 °C, VALUE = 65, THRESH = 0	Current temperature is within normal limits.
SMART overall-health	PASSED	The overall test did not detect a critical failure, but it does not guarantee the drive is fault-free.

Recommended actions

1. **Back up all important data immediately.**

The presence of pending and uncorrectable sectors indicates that data integrity could be at risk.

2. **Run a full surface scan (e.g., `smartctl -t long /dev/sdX`).**
   This will identify any additional bad sectors and confirm whether the current counts are increasing.
3. **Plan for replacement if the scan reports further errors or if the pending/uncorrectable counts grow.**
   Continuing to use the drive risks data loss.

Limitations

- The assessment is based on a single snapshot; we cannot determine a trend or predict when failure will occur.
- Vendor-specific encoding of raw values is not interpreted beyond the counts shown.
- The overall-health “PASSED” status does not negate the warning signs presented by the sector counts.
- No additional SMART attributes or error logs are available to refine the diagnosis.

Exhibit E-04

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
Critical Warning	0x00	No critical warnings present.
Temperature	35 Celsius	Temperature within normal operating range.
Available Spare	100%	Spare remains full; threshold not reached.
Available Spare Threshold	10%	Threshold value; current spare above threshold.
Percentage Used	3%	Only 3% of capacity used.
Power Cycles	150	Power cycles within typical limits for consumer SSDs.
Power On Hours	2400	2400 hours of operation; moderate usage.
Unsafe Shutdowns	0	No unsafe shutdowns recorded.
Media and Data Integrity Errors	0	No media or data integrity errors detected.
Error Information Log Entries	0	No error log entries present.
Warning Comp. Temperature Time	0	No warning temperature time recorded.
Critical Comp. Temperature Time	0	No critical temperature time recorded.

Recommended actions

1. Maintain regular backups of important data.

2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Limitations

- Single snapshot cannot establish trend or predict failure timing.
- No historical data to assess wear progression.
- Manufacturer-specific thresholds and encoding not provided.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-05

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
Critical Warning	0x00	No critical warnings indicated.
Temperature	35 Celsius	Within normal operating range.
Available Spare	100%	Spare capacity above threshold.
Available Spare Threshold	10%	Threshold value; spare above threshold.
Percentage Used	3%	Low endurance consumption.
Power Cycles	150	Neutral count; no indication of excessive cycling.
Power On Hours	2400	Neutral duration; no indication of excessive usage.
Unsafe Shutdowns	0	No unsafe shutdowns recorded.
Media and Data Integrity Errors	0	No integrity errors recorded.
Error Information Log Entries	0	No error log entries.
Warning Comp. Temperature Time	0	No warning temperature time recorded.
Critical Comp. Temperature Time	0	No critical temperature time recorded.

Recommended actions

1. Maintain regular backups of important data.
2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Limitations

- Single snapshot; cannot establish trend or predict failure timing.
- No vendor-specific documentation beyond provided thresholds.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-06

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD B
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: FAILED!
- available spare has fallen below threshold

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x01
Temperature:                        35 Celsius
Available Spare:                    5%
```

Analyze

Triage assessment

Monitor

The supplied evidence contains findings that warrant follow-up. A single snapshot does not establish a deterioration trend.

Evidence

attribute	reported_value	interpretation
Critical Warning	0x01	Flag indicates at least one critical condition, likely available spare below threshold.
Temperature	35 Celsius	Within normal operating range.
Available Spare	5%	Spare capacity below threshold of 10%.
Available Spare Threshold	10%	Threshold value for spare capacity.
Percentage Used	85%	High usage but not directly indicative of imminent failure.
Power Cycles	450	Moderate number of power cycles.
Power On Hours	12000	Moderate operational hours.
Unsafe Shutdowns	0	No unsafe shutdowns recorded.
Media and Data Integrity Errors	12	Count of integrity errors; may indicate data corruption.
Error Information Log Entries	12	Number of error log entries, matching integrity errors.
Warning Comp. Temperature Time	0	No time spent in warning temperature.
Critical Comp. Temperature Time	0	No time spent in critical temperature.

Recommended actions

1. Ensure important data is backed up before further investigation.
2. Compare with earlier reports, if available, and consult device documentation or qualified support about the flagged findings.

Limitations

- A single snapshot cannot establish a trend or predict failure timing.
- Vendor-specific interpretation of error counts is not provided.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-07

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD B
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: FAILED!
- available spare has fallen below threshold

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x01
Temperature:                        35 Celsius
Available Spare:                    4%
```

Analyze

Triage assessment

Urgent attention

The supplied evidence contains concerning indicators. Prioritize protecting important data; failure timing cannot be determined from this report.

Evidence

attribute	reported_value	interpretation
Critical Warning	0x01	Indicates available spare below threshold warning.
Temperature	35 Celsius	Current temperature; within typical operating range.
Available Spare	4%	Spare capacity below the supplied threshold of 10%.
Available Spare Threshold	10%	Threshold value for available spare.
Percentage Used	85%	Manufacturer-estimated endurance consumed; not occupied storage capacity.
Power Cycles	450	Number of power cycle events.
Power On Hours	12000	Total hours the drive has been powered on.
Unsafe Shutdowns	0	No unsafe shutdowns recorded.
Media and Data Integrity Errors	15	Count of media and data integrity errors.
Error Information Log Entries	15	Number of error log entries recorded.
Warning Comp. Temperature Time	0	No warning temperature time recorded.
Critical Comp. Temperature Time	0	No critical temperature time recorded.

Recommended actions

1. Prioritize protecting important data. If it is irreplaceable and the drive is malfunctioning, consult a qualified recovery specialist before further testing.

2. After protecting the data, seek device-appropriate diagnostics or qualified support to assess the concerning findings and whether replacement is warranted.

Limitations

- A single report cannot predict failure timing or establish when the recorded errors occurred.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-08

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
Critical Warning	0x00	No critical warnings indicated.
Temperature	35 Celsius	Within normal operating temperature range.
Available Spare	100%	Spare capacity well above the supplied threshold.
Available Spare Threshold	10%	Threshold value for spare capacity.
Percentage Used	3%	Manufacturer-estimated endurance consumed; low value.
Power Cycles	150	Neutral metric; no indication of abnormal usage.
Power On Hours	2400	Neutral metric; no indication of abnormal usage.
Unsafe Shutdowns	0	No unsafe shutdowns recorded.
Media and Data Integrity Errors	0	No media or data integrity errors recorded.
Error Information Log Entries	0	No error log entries recorded.
Warning Comp. Temperature Time	0	No warning temperature time recorded.
Critical Comp. Temperature Time	0	No critical temperature time recorded.

Recommended actions

1. Maintain regular backups of important data.

2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Important Notes

- Single snapshot cannot establish trend or predict failure timing.
- No vendor-specific thresholds beyond those supplied.
- No additional attributes or historical data available.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-09

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
SMART overall-health self-assessment test result	PASSED	No overall health warning.
Critical Warning	0x00	No critical warnings indicated.
Temperature	35 Celsius	Within normal operating range.
Available Spare	100%	Spare capacity above threshold.
Available Spare Threshold	10%	Threshold value for spare capacity.
Percentage Used	3%	Low endurance consumed.
Power Cycles	150	Neutral metric; no indication of abnormal usage.
Power On Hours	2400	Neutral metric; no indication of abnormal usage.
Unsafe Shutdowns	0	No unsafe shutdowns recorded.
Media and Data Integrity Errors	0	No media or data integrity errors.
Error Information Log Entries	0	No error log entries recorded.
Warning Comp. Temperature Time	0	No warning temperature time recorded.
Critical Comp. Temperature Time	0	No critical temperature time recorded.

Recommended actions

1. Maintain regular backups of important data.
2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Important Notes

- These fields do not predict remaining years or guarantee reliability.
- A single snapshot cannot establish a trend or predict failure timing.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-10

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
SMART overall-health self-assessment test result	PASSED	Overall health passed.
Critical Warning	0x00	No critical warning indicated.
Temperature	35 Celsius	Reported temperature in Celsius.
Available Spare	100%	Reported spare capacity is above the supplied 10% threshold.
Available Spare Threshold	10%	Supplied threshold value.
Percentage Used	3%	Manufacturer-estimated endurance consumed, not occupied storage capacity.
Power Cycles	150	Reported number of power cycle events.
Power On Hours	2400	Reported total powered-on hours.
Unsafe Shutdowns	0	Reported unsafe shutdown count.
Media and Data Integrity Errors	0	Reported media and data integrity error count.
Error Information Log Entries	0	Reported error log entry count.
Warning Comp. Temperature Time	0	Reported warning critical temperature time.
Critical Comp. Temperature Time	0	Reported critical temperature time.

Recommended actions

1. Maintain regular backups of important data.
2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Important Notes

- A single snapshot does not establish a trend or predict failure timing.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-11

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
SMART overall-health self-assessment test result	PASSED	Overall health assessment: PASSED.
Critical Warning	0x00	No critical warnings reported.
Temperature	35 Celsius	Reported temperature.
Available Spare	100%	Reported spare capacity.
Available Spare Threshold	10%	Supplied threshold.
Percentage Used	3%	Manufacturer-estimated endurance consumed.
Power Cycles	150	Reported number of power cycle events.
Power On Hours	2400	Reported total powered-on hours.
Unsafe Shutdowns	0	Reported unsafe shutdown count.
Media and Data Integrity Errors	0	Reported media and data integrity error count.
Error Information Log Entries	0	Reported error log entry count.
Warning Comp. Temperature Time	0	Reported warning temperature time.
Critical Comp. Temperature Time	0	Reported critical temperature time.

Recommended actions

1. Maintain regular backups of important data.
2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Important Notes

- These fields do not predict remaining years or guarantee reliability.
- A single snapshot does not establish a trend or predict failure timing.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-12

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
Critical Warning	0x00	No critical warnings.
Temperature	35 Celsius	Reported temperature.
Available Spare	100%	Reported spare capacity.
Available Spare Threshold	10%	Supplied threshold.
Percentage Used	3%	Manufacturer-estimated endurance consumed, not occupied storage capacity.
Power Cycles	150	Reported number of power cycle events.
Power On Hours	2400	Reported total powered-on hours.
Unsafe Shutdowns	0	Reported number of unsafe shutdown events.
Media and Data Integrity Errors	0	Reported media and data integrity error count.
Error Information Log Entries	0	Reported number of error log entries.
Warning Comp. Temperature Time	0	Reported warning critical temperature time.
Critical Comp. Temperature Time	0	Reported critical temperature time.

Recommended actions

1. Maintain regular backups of important data.
2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Important Notes

- These fields do not predict remaining years or guarantee reliability.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-13

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
smartctl 7.4
=== START OF INFORMATION SECTION ===
Model Number: SYNTHETIC NVME SSD A
NVMe Version: 1.4

=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

SMART/Health Information (NVMe Log 0x02)
Critical Warning:                   0x00
Temperature:                        35 Celsius
Available Spare:                    100%
Available Spare Threshold:          10%
```

Analyze

Triage assessment

No obvious warning in supplied data

No obvious warning was identified in the supplied measurements. This does not establish overall drive reliability.

Evidence

attribute	reported_value	interpretation
SMART overall-health self-assessment test result	PASSED	Overall health passed.
Critical Warning	0x00	No critical warning indicated.
Temperature	35 Celsius	Reported current temperature.
Available Spare	100%	Reported spare capacity is above the supplied 10% threshold.
Available Spare Threshold	10%	Threshold for spare capacity.
Percentage Used	3%	Manufacturer-estimated endurance consumed, not occupied storage capacity.
Power Cycles	150	Reported number of power cycle events.
Power On Hours	2400	Reported total powered-on hours.
Unsafe Shutdowns	0	Reported unsafe shutdown count.
Media and Data Integrity Errors	0	Reported media and data integrity error count.
Error Information Log Entries	0	Reported error information log entry count.
Warning Comp. Temperature Time	0	Reported warning-temperature time.
Critical Comp. Temperature Time	0	Reported critical-temperature time.

Recommended actions

1. Maintain regular backups of important data.
2. Compare future reports for changes; this snapshot is not a reliability guarantee.

Important Notes

- These fields do not predict remaining years or guarantee reliability.
- A single report cannot predict failure timing or establish when the recorded errors occurred.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-14

Lis Hadërgjonaj | B5KCCW

Original application output capture

Analysis settings

Temperature

0.20

Maximum output tokens

2048

S.M.A.R.T. Drive Triage

Paste a smartctl report to analyze potential drive problems.

AI-assisted triage, not a guaranteed failure prediction. Submitted text is sent to Groq. Remove serial numbers and other identifying information.

S.M.A.R.T. report

```
=== START OF INFORMATION SECTION ===
Device Model: SYNTHETIC SATA HDD C
Device type: ATA device

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

ID# ATTRIBUTE_NAME          FLAG   VALUE WORST THRESH TYPE      UPDATED WHEN_FAILED RAW_VALUE
  5 Reallocated_Sector_Ct   0x0033 100   100   010    Pre-fail  Always       -       0
  9 Power_On_Hours          0x0032 095   095   000    Old_age   Always       -       3200
197 Current_Pending_Sector  0x0012 100   100   000    Old_age   Always       -       6
198 Offline_Uncorrectable   0x0010 100   100   000    Old_age   Offline      -       6

No previous reports or self-test results are supplied.
```

Analyze

Triage assessment

Urgent attention

The supplied evidence contains concerning indicators. Prioritize protecting important data; failure timing cannot be determined from this report.

Evidence

attribute	reported_value	interpretation
Reallocated_Sector_Ct	RAW_VALUE=0	Reported reallocated sectors.
Power_On_Hours	RAW_VALUE=3200	Reported total powered-on hours.
Current_Pending_Sector	RAW_VALUE=6	Reported pending sectors.
Offline_Uncorrectable	RAW_VALUE=6	Reported uncorrectable sectors; these may overlap with the pending sectors.

Recommended actions

1. Prioritize protecting important data. If it is irreplaceable and the drive is malfunctioning, consult a qualified recovery specialist before further testing.

2. After protecting the data, seek device-appropriate diagnostics or qualified support to assess the concerning findings and whether replacement is warranted.

Important Notes

- A single snapshot does not establish a trend or predict failure timing.

This AI assessment can be wrong and cannot guarantee drive reliability.

Exhibit E-15 — Missing telemetry regression

Lis Hadërgjonaj | B5KCCW

Triage assessment

Insufficient information

The supplied information is insufficient for a reliable drive-health assessment.

Evidence

No usable measurements were extracted.

Recommended actions

1. Obtain a complete readable S.M.A.R.T. report and the drive model, removing identifying details before submission.
2. For unclear raw values or unavailable telemetry, consult device or adapter documentation or vendor support.

Important Notes

- Missing telemetry alone does not establish a hardware fault.

This AI assessment can be wrong and cannot guarantee drive reliability.